MILITARY SPECIFICATION

TRUCKS, LIFT, FORK, DIESEL--ENGINE-DRIVEN:

PNEUMATIC-TIRED, ROUGH TERRAIN;

6,000 POUND CAPACITY

AT 24-INCH LOAD CENTER

This amendment forms a part of Military Specification MIL-T-53020, dated 14 March 1983, and is approved for use by the US Army Mobility Equipment Research and Development Command, Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 2

2.1.1, under SPECIFICATIONS, MILITARY, add:

MIL-T-62314 Test Equipment (Simplified) for Internal Combustion Engine (STE/ICE)."

PAGE 4

2.1.2, add:

"TACOM DOCUMENTS AND DRAWINGS

Report No. CR-82-588-003 - STE/ICE Design Guide for Vehicle Diagnostic Connector Assemblies.

DWG 12258941 - Connector, Receptacle 54 Pin.
DWG 12258955 - Vehicle Test Card.

(Application for copies should be addressed to Department of the Army, U.S. Army Tank-Automotive Command, ATTN: DRSTA-RGD, Warren, MI 48090.)"

PAGE 7

3.1.3.3, line 2, between "railroad" and "and" insert "in accordance with MIL-HDBK-157".

3.1.3.4, line 2, delete "roll-on barge" and add "roll-on/roll-off (RORO) ships, C-8 and larger, lighter aboard ship (LASH), barge".

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* 3.5.1, line 3, after "SAE J231", insert "if necessary for transportability, the ROPS and FOPS shall be removable".

3.5.1, line 4, delete "eyes" and add "eyes/devices".

3.5.1, add new sentence at end of paragraph, "Quick disconnects for electrical lines shall be provided to facilitate removal of the ROPS and FOPS."

PAGE 11

3.9.1.1, line 1, delete "(modified front end loader)".

3.9.1.1.1, delete in its entirety.

PAGE 13

3.9.11, line 3, after "... plane of the front tires" delete remainder of the paragraph and substitute "when the forks are 100 inches ±2 inches above ground. With the forks horizontal and at a fork height of 93 +1/2 inches and the front tires in contact with a 90 inch high vertical wall, no other part of the fork--lift shall contact the wall."

* 3.9.11, delete entirely and insert:

"3.9.11 Fork reach. The forklift shall be capable of engaging and lifting and disengaging the rated load when the heel, front face, of the fork is at a minimum of 48 inches forward of the vertical plane of the front tires when the forks are 100 inches +2 inches above the ground."

PAGE 14

3.9.16, line 5, delete "85" and substitute "83".

3.9.18, line 5, delete "and" and add "or".

PAGE 19

3.11.1, at the end of paragraph add:

"The transmission shall be provided with a disconnect control. The disconnect control shall disengage the transmission and apply the service brakes while allowing the engine to be operated throughout the entire speed range. The forklift shall be capable of achieving specified lifting, lowering, and tilting rates of the rated load while the disconnect control is engaged and transmission control is in forward position, without moving the truck."

3.11.1.1, delete the fourth, fifth, and sixth sentences.

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3.12.1.4, lines 9 and 10, after "saddle worm thread type with stainless steel inner liner" add "or a heavy duty T-bolt clamp".

3.12.2 Delete in its entirety and add:

"3.12.2 Hydraulic pumps. The hydraulic pump(s) shall have been tested in accordance with NFPA Recommended Standard T.3.9.18-1976. When tested in this manner, the contaminant tolerance profile for the hydraulic pump(s) shall either:

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a. Not fall below or cross the Bl0 curve of the hydraulic system filter when tested in accordance with MIL-F-52723 (see 4.6.2.21.2).

or else

b. The pump Omega rating shall be less than the filter Beta 10 rating."

PAGE 2.3

3.12.4, line 2, change "screen cap" to "screen, cap".

PAGE 24

* 3.13, lines 17 and 18, delete "Lift arm . . . SAE J38." and substitute:

"Lift arm safety device(s) shall be furnished. The design of the lift arm safety device(s) shall be in accordance with SAE J38.

PAGE 26

* 3.15.2, line 1 after "wide base," add "or 65 series,". Line 4, delete "table WB-5 of" and after "the TRA Yearbook" add "and the tire manufacturer's published literature for 5 mph load ratings."

3.17.2, line 2, after "engine failure" acid "or steering power source failure".

PAGE 27

3.18.2, line 8, delete "70" and add "50"; line 9, delete "90" and add "75".

PAGE 28

3.21, line 6, after "one person" add "wearing gloves".

3.21, at end of last sentence change "opened." to "opened/removed."

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3.22, line 3, delete “incline." and add "incline at a minimum of 5 degrees to a maximum of 80 degrees .“

PAGE 29

3.23.6, line 2, delete “hours shall” and add “hours up to a minimum of 9999 hours shall”.

PAGE 30

* 3.23.9, delete the last sentence and add new sentences:

“All internal and external lights and horn shall be made inoperable when blackout lights are used (see 3.24.4.1) except forewarning lights, such as temperature and oil lights, which shall remain operative in the blackout mode. The blackout switch(es) shall have an interlock device to prevent the accidental energizing of any part of the normal lighting system when in the blackout mode.”

3.23.10, line 1, change “pedal(s)” to “pedals"; delete, “A 1 or 2 pedal system is permitted.” and substitute “A 2 pedal system is required.”; and line 2, delete, “For the 2 pedal system,".

* 3.23.10, lines 2 and 3, change “... for convenient right foot operation . ..’1 to read "... for either convenient right or left foot operation . ..”.

* 3.23.10, lines 9 and 10 delete sentence “For the 1 pedal system, ... disconnect and braking pedal.”

3.23.12, after first sentence add “Forward, reverse, and speed ranges shall be labeled in accordance with SAE J298."

3.23.13, line 8, delete "pounds at” and add "pounds and shall have a minimum force of 2 pounds at".

PAGE 31

3.23.14, line 1, after “Starter”, add “switch”,

3.23.14, line 3 after "starter", add "switch”.

PAGE 32

3.24.4.1, line 1, delete “shall” and add "may".

PAGE 33

* 3.26, add “assembly" after “Pintle hook".

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* 3.26.1, line 1, add “assembly” after “Pintle hook". Lines 1, 2, 4 and 5 add “assembly" after “pintle hook". Add new sentence after second sentence: “The pintle hook assembly shall be capable of being rotated 360 degrees about the horizontal axis.”

3.27, line 9, delete “on the other”, and add "on another".

3.27, line 10, add new sentence “All doors and windows shall be capable of being opened and closed by personnel wearing gloves."

* 3.27, line 10, revise sentence to read “All doors and windows that can be opened shall be capable of being opened and closed by personnel wearing gloves.

* 3.27, line 10, after “catches” add “except that cab door(s) need not latch open if the door is equipped with a window that can be latched open to provide adequate operator ventilation”.

PAGE 34

3.27.4, at end of paragraph add “If a fuel-fired heater is furnished, it shall be in accordance with SAE JI024.”

3.29, last sentence, change “A lubrication data plate in accordance . ..” to “A lubrication data plate or commercially proven decals in accordance . ..".

3.29.1, at end of paragraph add ’’Accessibility shall be in accordance with SAE J925 for personnel normally clothed.”

PAGE 36

3.32.1, line 1, delete “plates.“ and add "plates/labels.”

3.32.1, line 2, delete “plates’s and add “plates/labels”.

3.33, after first sentence add the following:

"Parts and components maybe primed with alkyd base primer (color immaterial) which has ‘been allowed to cure for a minimum of 14 days and then cleaned and painted as specified herein either prior to or at final truck assembly painting."

3*35, line 4, after "fastener” add “accessible and operable by personnel wearing gloves.”

3.36, delete “Decontamination unit” and add “Decontamination bracket assembly".

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3.40, add "e. STE/ICE for test in 4.6.2.41.1.1'

* 3.40 add 'if. DCA tester for use in 4.6.2.41."

After 3.40, add the following:

"3.41 Diagnostic connector assembly (DCA) measurement capabilities. Unless otherwise specified, see 6.2, the forklift will be furnished with the following DCA measurement capabilities:

Engine RPM (Average)	Battery Voltage
*power Test (RPM/Sec)	Starter Motor Voltage
*Compression Unbalance	Starter Negative Gable Voltage Drop
Fuel Supply Pressure	*Starter Current First Peak
Fuel Return Pressure	*Internal Battery Resistance
Fuel Filter Pressure Drop	*Battery Resistance Change
Fuel Solenoid Voltage	Alt/Gen Output Voltage
Engine Oil Pressure	Alt/Gen Field Voltage
Engine Oil Filter Pressure Drop	Alt/Gen Neg Cable Voltage Drop
Engine Coolant Temperature	*Starter Circuit Resistance
Starter Solenoid Voltage	Alternator Output Current Sense
*Starter Current (Average)	Alternator A/C Voltage Sense
Hydraulic System Pressure	Transmission Oil Pressure

All requirements for DCA shall be in accordance with the STE/ICE design guide for vehicle Diagnostic Connector Assemblies, Report No CR-82-588-003. The DCA itself shall be in accordance with TACOM drawing No 12258941 and shall interface with the Simplified Test Equipment/Internal Combustion Engine (STE/ICE) as specified in MIL-T-62314(AT) Appendix B. A separate wiring harness shall be provided for the DCA and shall include all wiring and necessary hardware to perform required capabilities. The contractor shall also provide STE/ICE vehicle test cards as specified on TACOM drawing No. 12258955 and incorporate the test cards into the forklift technical manuals being furnished. Values for test identified by an asterisk above will be provided by the Government for incorporating into the Test Card and manuals. All other values to be determined by the contractor."

* 3.41, delete the following from the list, "Hydraulic System Pressure," "Alternator A/C Voltage Sense", and "Alternator Output Current Sense."

4.3.2, fifth sentence, delete in its entirety, and add "Failure of any test, except for EMP testing as specified in 4.6.2.40.l shall be cause for rejection."

4.5.3.3, line 3, delete "and 4.6.2.18 (test method 17 only)," and substitute "4.6.2.18 (test method 17 only), and 4.6.2.41.1."

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PAGE 43

TABLE III, line 3, delete Defect 113, in its entirety.

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TABLE III, line 25, Defect 150, delete “X” in columns 1, 2, and 3.

PAGE 47

TABLE III, add the following:

Defect, "219. DCA not as specified." “X” in columns 1, 2, and 3. “3.41" in column 5.

PAGE 48

TABLE IV, line 20, delete, "Overload sensor ... only)." completely.

PAGE 49

TABLE IV, line 5, "Service brake" delete “X" in column 4. Line 7, "0ther transportability'*, add "3.1.3.3” to column 7.. Line 13, “Leakage", delete “X" in column 4.

PAGE 50

TABLE IV, add ‘'X” in columns 1, 2, 3, and 4. Add “Diagnostic connector assembly capability.” in column 5. Add “4.6.2.41” in column 6. Add “3.41” in column 7. Add "X" in columns 1, 2, and 3. Add “-" in column 4. Add "DCA-STE/ICE interface." in column 5. Add “4.6.2.41.l" in column 6. Add "3.41” in column 7.

PAGE 51

4.6.2.11.1, delete in its entirety.

PAGE 52

4.6.2.20, line 5, delete “3.1.3.2, and" and add “3.1.3.2, 3.1.3.3, and”.

PAGE 57

4.6.2,40.1, second sentence, delete end of sentence following “Government facility.”

* 4.6.2.40.1, line 4, between “testing" and "at" insert “by the Government."

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After 4.6.2 .40.1 add:

"4.6.2.41 DCA capability The contractor will performs 100 percent visual inspection of all DCA wiring harnesses prior to installation of the forklift to insure wire and connection are free of error. After installation on the forklift, test the DCA harness (reference chapter 5 of design guide for DCA) On each vehicle with a DCA tester prior to interfacing with STE/ICE. Nonconformance to 3.41 shall constitute failure of the DCA.

* 4.6.2.41, revise the first and second sentences as follows:

"The contractor will perform a 100 percent visual inspection of all DCA wiring harnesses prior to installation on the forklift to in-sure wires and connections are free of error. After installation on the forklift, test the DCA harness (reference chapter 5 of design guide for DCA) on each vehicle with the DCA tester (GFE) prior to interfacing with STE/ICE as required under 4.6.2.4.41.1.[8]'

"4.6.2.41.1 DCA - STE/ICE interfaces. Upon completion of inspection and tests specified in 4.6.2.41, test the DCA for all measurements specified in 3.41 using STE/ICE. Nonconformance to 3.41 shall constitute failure of DCA."

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6.2, add "1. When DCA is not required (3.41)."

6.3, second sentence, delete "first-produced", and add "first article (reproduction)".

PAGE 61

Test Method No. 1, paragraph 2., add "d. 90 inch high wall."
Under paragraph 3., c., line 1, delete "forks to" and add "forks (with load) to". Under paragraph 3., c., delete last sentence.

Add new paragraph "d. Raise forks (without load) to 93 inches (+1/2 inch) above the ground. Approach the 90 inch high wall until the front tires come into contact with the wall. Other than the front tires, note if any part of the forklift contacts the wall."

*Test Method No. 1, delete paragraphs 2., d., 3., d.

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Test Method No. 5, first sentence of sub-paragraph 3., c., (3), delete "full forward 10°" and add "forward 10 degrees".

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Test Method No. 31, 4 a., add new sentence at end of paragraph, "Block thermostat in open position."

PAGE 9$

Delete Test Method No. 33.

The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the Previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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